

Vedant Rupwal

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Professional Summary

Math and Computer Science student with a strong foundation in computational modeling, machine learning, and spatiotemporal data analysis. Experienced in migrating complex mathematical algorithms to Python and developing geospatial features for consumer-facing applications. Seeking to leverage expertise in Data Science and AI to deliver high-impact, data-driven solutions.

Education

University of Illinois at Chicago | Chicago, IL

- Bachelor of Science in Math and Computer Science | 2023 - 2026
- Minor in Economics | 2023 - 2026

Experience

AI & Data Advisor | Eigen Axis

Dec 2025 - Present

- **Systems Architecture:** Architected a production-ready, full-stack web application and Retrieval-Augmented Generation (RAG) backend tailored for clinical knowledge discovery and managing LLM training pipelines.
- **Data Engineering:** Engineered an automated, multi-channel data ingestion pipeline with exponential jittered backoff and a Dead Letter Queue (DLQ), reliably extracting structured texts from Wikipedia APIs, PubMed, and local PDFs.
- **Search Infrastructure:** Developed a hybrid embedding engine combining dense neural vectors (Sentence-Transformers) and statistical BM25 sparse matrices, streaming real-time indexing into a localized Qdrant vector database.
- **Model Deployment:** Integrated PyTorch and TensorFlow training loops with a FastAPI backend and WebSockets, enabling real-time monitoring of model states (Idle, Training, Running) and seamless multi-environment deployments.
- **Security & Compliance:** Built a highly secure authentication domain with TOTP MFA, PBKDF2-HMAC-SHA256 password hashing, and Role-Based Access Control (RBAC) to govern schema-driven YAML configurations and track system audit logs.

Data Science Research Assistant | University of Illinois, Chicago

Jan 2025 - May 2026

- **Algorithm Migration:** Pioneering the migration of complex MATLAB research codebases to Python, improving the scalability and efficiency of data processing pipelines, and improving processing time by 98% (from 3 months to 30 hours).
- **Spatiotemporal Analysis:** Performing advanced spatiotemporal mapping of topography to analyze the thickness of transition in ultrathin foam films.
- **Statistical Modeling:** Analyzing raw data to derive meaningful insights and visualize physical properties using mathematical modeling.

Personal Project

Ask the Pandit | Full-Stack RAG Developer

Tech Stack: Python, JavaScript, HTML/CSS, FastAPI, ChromaDB, Hugging Face, OpenAI SDK, Qwen 2.5

- End-to-End Application: Developed a context-aware Chrome extension and highly optimized FastAPI backend to deliver precise, text-backed RAG answers sourced from extensive theological scriptures.
- Database Partitioning: Architected a dynamic hybrid database routing system using ChromaDB to aggressively split massive datasets, optimizing system memory limits and accelerating semantic search speed.
- Advanced Retrieval: Implemented a two-stage retrieval pipeline utilizing all-MiniLM-L6-v2 for rapid similarity search and ms-marco-MiniLM-L-6-v2 CrossEncoders to rerank top vectors for deep reading comprehension accurately.
- Dynamic UI Integrations: Built a Shadow DOM-injected interface that intelligently scrapes active webpage elements, feeding up to 6,000 characters of real-time contextual reading material into the LLM.
- Safety Engineering: Designed strict, prompt engineering guardrails, including zero hallucination locks and false-premise handlers, to restrict AI responses to verified database texts and correct logical traps.

Econ Predictor | Lead AI & Simulation Architect

Tech Stack: Python, Next.js, FastAPI, JAX, EconML, DoWhy, Ray/RLlib, Qdrant, Docker

- System Architecture: Architected an end-to-end, 10-phase systemic fragility and economic-risk prediction pipeline, integrating a deterministic Python backend with a Next.js/React executive wargaming console.
- Causal Inference & Math Engines: Engineered structural causal models (DAGs) using EconML and DoWhy, alongside Extreme Value Theory (EVT) and GARCH-style mathematical transforms to detect volatility regimes and hidden structural confounding.
- Agent-Based Simulation: Developed a GPU-accelerated agent-based simulation utilizing JAX and sparse network topologies (CSR-like arrays) to model financial contagion, prospect-theory utility, and phase-transition stability across millions of simulated nodes.
- Data Engineering: Built automated, schema-enforced ingestion pipelines (Pydantic) to extract and align macroeconomic datasets (FRED, Zillow, CFPB) into point-in-time feature snapshots for dynamic stress testing.
- LLM & RAG Integration: Integrated a local vLLM and Qdrant vector database to generate strictly parsed, zero-hallucination after-action reports and wargame strategy recommendations.

Certifications

- Probabilistic Graphical Models – Stanford University (Jun 2026)
- MLOps | Machine Learning Operations – Duke University (Jun 2026)
- Microsoft AI & ML Engineering Professional Certificate – Microsoft (Jun 2026)
- Google Advanced Data Analytics Professional Certificate – Google (Jun 2026)
- Deep Learning using TensorFlow – IBM (Jun 2025)
- Accelerated Deep Learning with GPU – IBM (Jun 2025)

Technical Skills

- Artificial Intelligence & ML: Machine Learning Models, Bayesian Networks, Markov Models, GANs
- Cloud & MLOps: AWS SageMaker, Microsoft Azure, MLflow, DevOps, Azure DevOps
- Data Science: Spatiotemporal Mapping, Statistical Analysis, Data Visualization, Python, R, MATLAB
- Tools & Infrastructure: Linux, SQL Server, Figma, Big Data Warehousing
- Languages: Python, C/C++, Java, R, MATLAB